Agricultural Systems 227 (2025) 104364

Contents lists available at ScienceDirect

Agricultural Systems

journal homepage: www.elsevier.com/locate/agsy

Exploring opportunities to improve crop-livestock integration and production in mixed farms with a serious game: The case of semi-arid Burkina Faso

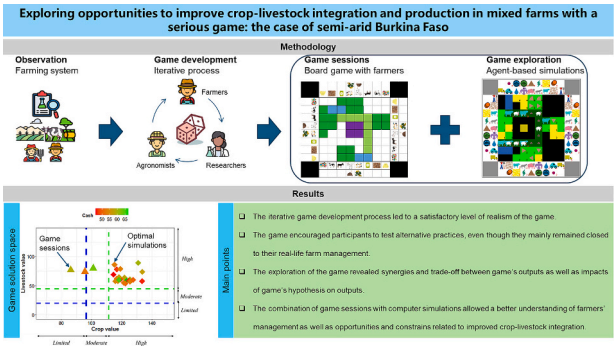
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HIGHLIGHTS

- A serious game (QUEEN) was developed to explore options for stronger crop-livestock integration and improved production.
- Exploring QUEEN's solution space helped understand outcomes linked to game mechanics and underlying hypotheses.
- QUEEN has a satisfactory level of realism, and motivated participants to test alternative farm practices.
- Combining game sessions and simulations clarified opportunities and constraints in crop-livestock integration.
- Serious games are relevant tools in the process of co-designing improved mixed crop-livestock farming systems.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: David Harris

Keywords:
Serious game
Co-learning
Participatory research
Solution space

ABSTRACT

Context: Serious games are increasingly recognised as powerful tools to deepen knowledge and stimulate (social) learning among game participants.
Objective: We developed a serious game, called QUEEN, to better understand farmer decision-making and to explore potential options to improve crop-livestock integration and production as seen from the farmer perspective using a case study of a semi-arid farming system in Burkina Faso.
Methods: The development of the serious game was an iterative process with several rounds in which local farmers and local researchers played a pivotal role. The game included options for soil fertility management, crop

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https://doi.org/10.1016/j.agsy.2025.104364
Received 22 October 2024; Received in revised form 18 April 2025; Accepted 24 April 2025
Available online 2 May 2025
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Legumes
Sub-Saharan Africa

and livestock production, market opportunities and interactions among participants. Three game sessions were organized to collect data on farmers' decision-making and management strategies. Beyond gameplay, the game was used as an experimental platform—its solution space was explored through simulations to assess the outcomes of possible strategies, and the impacts of assumptions and game's mechanics on the game's outcomes.
Results and conclusions: We showed that while the participants remained generally close to their real-life farm management, in the game they all applied mulch, cereal-legume strip intercropping and rotations, which is less common in reality. This demonstrates the capacity of the game to encourage farmers to explore alternative practices. The solution space exploration helped identify trade-offs between objectives (e.g., livestock production vs. income generation), and pathways to improve crop-livestock integration.
Significance: QUEEN is a valuable learning and research tool for Burkina Faso and other semi-arid systems, engaging stakeholders in critical reflection on system constraints and innovations.

1. Introduction

The semi-arid zone of West-Africa is mainly dependent on mixed crop-livestock farming systems to generate food and income (Giller et al., 2021). These mixed crop-livestock farming systems are rather in a vicious cycle than in a virtuous one (Henderson et al., 2016). The productivity of the cropping system and the livestock system remains far from their ecological potential largely due to limited application of external and internal input, such as concentrate feed, nutrients and irrigation (FAOSTAT, 2023; Svendsen et al., 2009). Also the potential benefits of nutrient recycling remain largely underexploited (Grillot et al., 2018; Rufino et al., 2007). As a result, these farming systems heavily rely on external inputs (e.g. fodder and grain) to at least meet livestock feed and human food requirements (Assogba et al., 2023).

In semi-arid Burkina Faso (West-Africa), food security is a major concern (Fraval et al., 2020). Research efforts to sustain food security target several interconnected components (e.g. crops and soil) and levels of farming systems with interventions such as: (1) breeding for improved crops and livestock (Ouédraogo et al., 2021; Vom Brocke et al., 2020) at plant/animal level, (2) soil fertility management to support higher yields (Adam et al., 2020; Garba et al., 2018) at plot and farm level, (3) spatial and temporal cropping system management (Bayala et al., 2012; Sanou et al., 2016) to enhance resources (e.g. water) use efficiency at plot and farm level, and (4) reorganization of space and management of common resources to improve nutrients recycling at village level (Berre et al., 2021). Consequently, enhancing farming systems to sustain food security can benefit from an integrated approach that incorporates insights from multiple system components across different levels (Descheemaeker et al., 2018). Practically, identifying suitable interventions involves selecting ensembles of options that farmers can easily adopt and adapt (Périnelle et al., 2024; Ronner et al., 2021). This is easier said than done as smallholders farmers in sub-Saharan Africa (SSA) are often operating with little room to manoeuver given their socio-economic constraints (Ogisi and Begho, 2023). In this context, participatory research that includes farmers from design to assessment of options has been effective to co-design tailored and relevant options (Descheemaeker et al., 2019; Falconnier et al., 2017; Périnelle et al., 2024).

As a participatory approach, serious games are gaining popularity (Den Haan and Van der Voort, 2018) as a tool for co-learning and co-design because of their ability to represent complex systems in a way that is understandable for both researchers, farmers and other stakeholders (Moojen et al., 2022). In the context of farming systems, serious games have been proven useful to stimulate farmers and researchers to actively discuss their understanding of systems and proposed options for improvement (Speelman et al., 2014). Serious games that mimic complex systems in a stylized way can function as boundary objects (e.g. game board, pawns and rules) that facilitate reflection and discussion among game participants (van Noordwijk et al., 2020). As a stylized representation of reality, serious games can be seen as models (Valkering et al., 2009) that can be helpful for fostering discussion between farmers and researchers and co-creating knowledge (e.g. on water-related risks/conflicts management) that is useful for both. In serious games, complex mechanisms are converted into rules that help to illustrate the processes leading to the current system state. By serving as a support for learning and discussing (Rodela et al., 2019), serious games offer the opportunity to explore and design alternative management options to help in the resolution of issues identified by multiple stakeholders. Serious games can be used to explore the potential effects of these options on the components of the farming system (Andreotti et al., 2020; Ryschawy et al., 2022). Some serious games have been developed in SSA on various subjects, such as agriculture (Pfeifer et al., 2021), land use (Michalscheck et al., 2020), and water management (Mossoux et al., 2016; Wamucii et al., 2024), but their development and use in SSA remain limited (Flood et al., 2018). To the best of our knowledge, there is no serious game developed and applied in semi-arid West-Africa that offers a holistic representation of mixed crop-livestock farms. Furthermore, serious game exploration allowing to better understand probabilities of serious games outcomes and how to reach these outcomes does not currently receive enough attention from researchers (Sari et al., 2024). Whereas game exploration offers opportunities to discuss system management options with participants, it can also help researchers to better reflect on the influence of hypotheses underlying game development on game outcomes. Moreover, the analysis of pathways to achieve outcomes can provide insights on the realism of the serious game based on participants' and researchers' knowledge of the system.

In this paper, we developed a novel serious game aiming at understanding farmers' decision-making and exploring options for stronger crop-livestock integration and improved production of each sub-system (crop and livestock). The objectives of the game were (1) to understand farmers' decision-making regarding their resources (land, livestock, crop, money) in the game sessions, (2) to facilitate communication among farmers, and among farmers and researchers, (3) to serve as a virtual laboratory for farmers and researchers to explore the effects of soil fertility, crop or livestock management options at farm scale and (4) to determine with farmers the limits and opportunities for improvement of the farming system performance. The objective of each participant in the game was to manage their farm and increase farm-level crop and livestock production and income as much as possible. The game integrates biophysical, economic and social dimensions of the farming system. We analyzed the degree of realism of the game and participants' choices in the game using a post-game questionnaire. Moreover, the solution space (all the possible outcomes) of the game was explored with computer simulations and farmers' outcomes were analyzed in comparison to theoretical optimal game outcomes.

2. Materials and methods

2.1. Study area

The serious game entitled QualitativE Evaluator of farm productioN (QUEEN) was developed and applied in semi-arid Burkina Faso, in two villages: Yilou and Tansin (Fig. 1). Yilou and Tansin are located in the provinces of Bam and Sanmatenga respectively. The local language spoken is Mooré. The climate in the study area is characterized by one rainy season (June–October) and one dry season (November–May). The

average annual rainfall is 676 mm, the average maximum temperature is 39 °C in April and the average minimum temperature is 18 °C in January. The farming system is dominated by mixed crop-livestock farms. The main cultivated crops are sorghum, millet, cowpea, peanut and sesame. Four farm types have been identified in the study area (Assogba et al., 2022) and included subsistence-oriented crop (SOC), subsistence-oriented livestock (SOL), market-oriented and diversified (MOD) and land constrained and livestock (LCL) farms. SOC and SOL farms are the dominant farm types in the area. These farm types were characterized by the smallest family size and revenue and differed mainly in the intensity of crop and livestock production (Table S1, supplementary materials). MOD farms had a bigger herd size than SOC and SOL as well as the highest revenue with more than half of its revenue coming from off-farm activities. LCL farms, only present in Yilou, were agro-pastoralists which had by far the largest herd size dominated by cattle. They cultivated the smallest area and relied mainly on grazing on rangelands in the village and transhumance (migration southward to get more abundant forage) to meet their livestock feed requirements.

The present study was part of the 3F (Feed the soil and Feed the cow to Feed the people) project that was implemented in Yilou and Tansin. Part of the project consisted of diversification of the sorghum-based system, using legumes in sole cropping, strip intercropping and rotation. The diversification options introduced by the 3F project were included in the game. The farmers who participated in game development (testing and game sessions) previously participated during three years in village demonstration trials on strip intercropping of sorghum with cowpea and mungbean as well as sorghum-cowpea/mungbean rotation. Mungbean is not commonly grown in the study area but has been recently introduced and used in on-farm participatory trials (Raboin et al., 2023).

2.2. Game development

The development of the serious game QUEEN (Fig. 2) started with the ARDI (Actors, Resources, Dynamics and Interactions) approach (Etienne et al., 2011) to identify the components of a socio-ecological system (here the farming system) and the interactions between the components (here the farms) through time (2.2.3). The ARDI approach

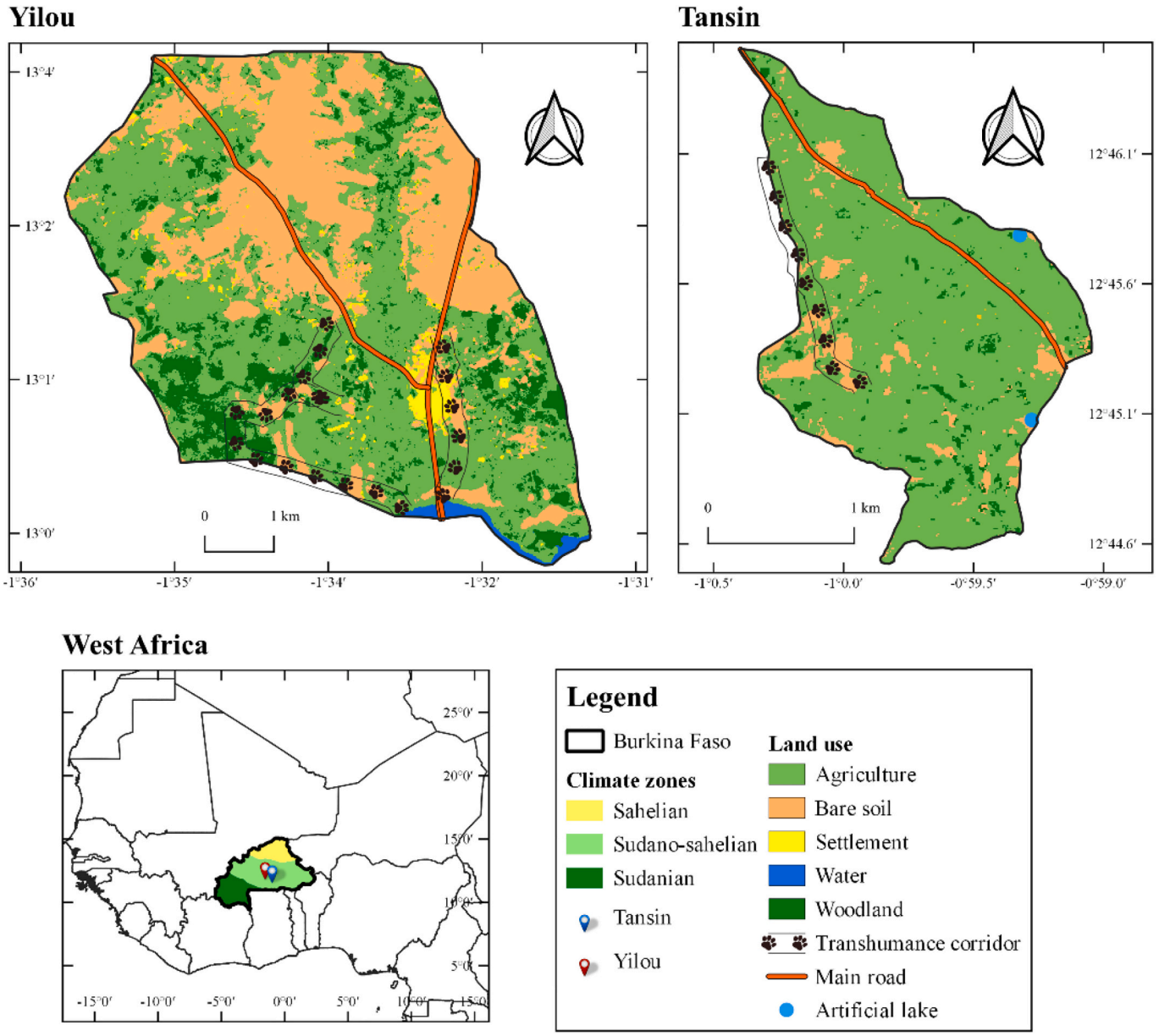


Fig. 1. map of the study area.



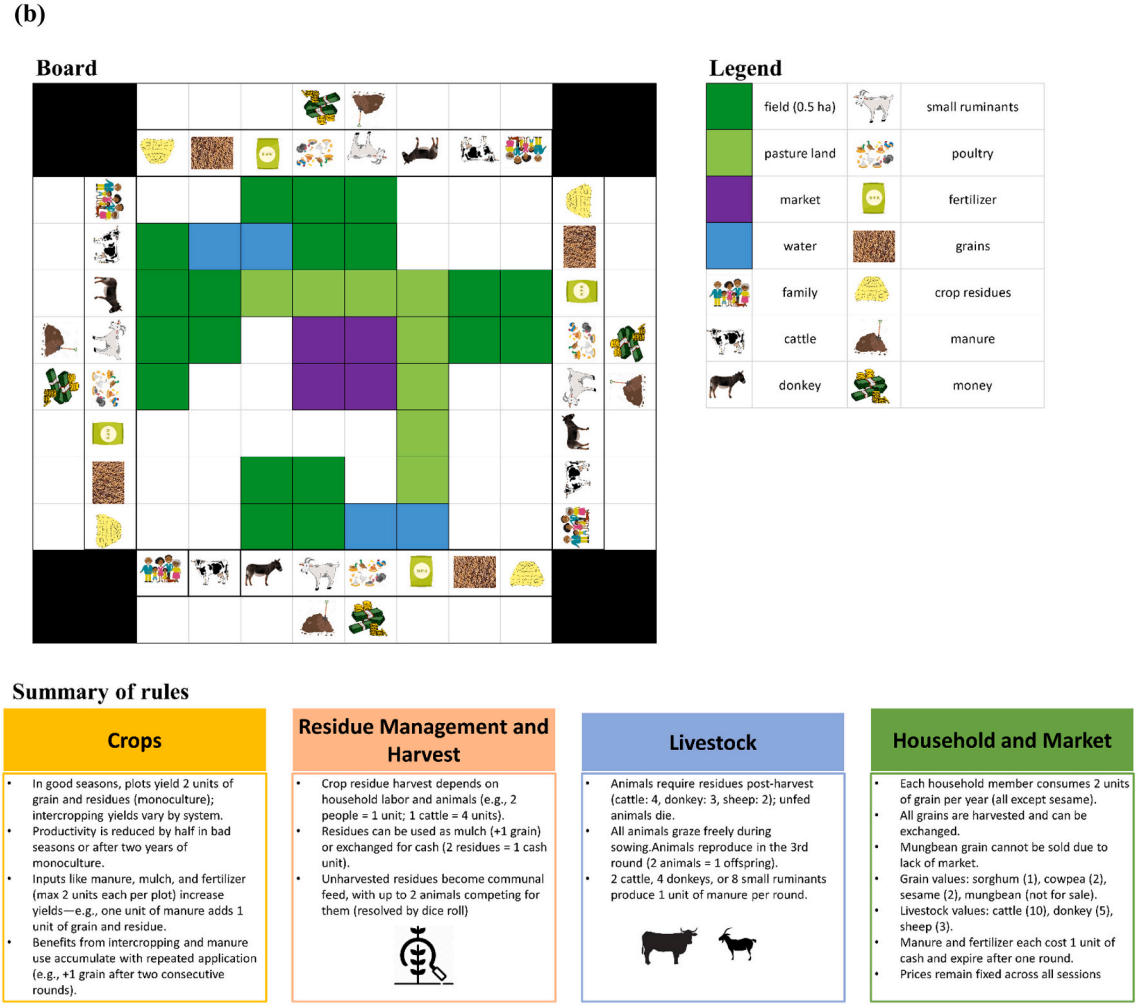


Fig. 2. Photo of a game session showing the participants around the game board (a). Board layout and summary of rules of the serious game QUEEN (QualitativE Evaluator of farm productioN) (b).

allows systems thinking by making explicit the complexity of interactions in socio-ecological systems. The approach can help determine key interactions to include in serious games (or other models) and ground serious games in the real-world context (Sari et al., 2024). The ARDI was implemented mainly based on the findings from previous participatory research on farm system characterization in our study area (Assogba et al., 2023; Assogba et al., 2022). Additionally, relevant literature on socio-ecological aspects of farming systems in semi-arid West-Africa (2.2.3) were also included in the ARDI to develop the mechanics of the game (2.2.4).

2.2.1. System description

The farming system components and relations that served as basis to the development of QUEEN included the four farm types present in the study area (Fig. S1). In each farm, households managed their livestock and fields to get grains, crop residues, manure and cash. Field management included the choice of crops, and the way they are managed (manure/mulch/fertilizer inputs, intercropped or not, in rotation or not), as well as crop residues harvesting. Livestock were fed with crop residues that were either harvested or left on field and forage in grazing areas. Several direct exchanges among households and with the market could occur. Farm resources that can be directly exchanged between households and in the market included manure, fertilizer, grains, crop residues, livestock and cash. Common resources considered in the farming system were crop residues left on the field after harvest and forage from grazing areas. Every animal in the system could graze on crop residue left on fields or in grazing areas. The climate and the market were the only two external drivers considered. Climate influenced the system through rainfall which directly impacts crop production (see Box S1). Households could buy and sell resources (2.2.3) from the market, which we assumed to be stable and unlimited, meaning that it was possible to sell and buy goods at constant prices.

2.2.2. Game development and testing

The development and testing of QUEEN was an iterative process involving researchers and farmers. Following the identification of actors, resources, dynamics and interactions to be included in the game, a prototype (first version) of the game was developed. The prototype of QUEEN was evaluated by ten researchers familiar with the farming system in several test sessions of the game (Table 1). The objectives of the test sessions with researchers were to assess the ease to play the game as well as its adequacy with regards to the objectives of the game. Based on these first tests, the mechanics of the game was adjusted before playing it with eight volunteer farmers (four in each village). After the test sessions with farmers, they were asked to provide feedback on the ease to play and the realism of the game as well as advice on how to improve it. Based on famers' feedback, the game was updated before testing it again with researchers and farmers. The whole adjustment process leading up to the final version lasted nine months (Table 1).

In its final version, QUEEN placed participants in the role of smallholder farmers managing crops, livestock, and household needs in a semi-arid environment (Fig. 2). The game included several rounds and each round represented a farming season with changing weather conditions that affect productivity (2.2.4). Participants made strategic decisions on crop choice, intercropping, input use (manure, fertilizer, mulch), and residue management. Livestock provided manure but also required feed, creating trade-offs. Crop residues not harvested can be shared or competed for, adding a social dimension to the game. The game allowed for self-organization and interactions among players.

2.2.3. ARDI of the farming system and its implementation in QUEEN

2.2.3.1. Actors. QUEEN was designed with four participants (players) representing the four farm types of the local farming system. Participants' roles were based on the farm types identified earlier in the study area.

2.2.3.2. Resources. The resources of the farm system included the cultivated fields, grains, livestock (cow, donkey and small ruminants), manure, and mineral fertilizer. In addition, the game included representations of crop residues, forage in grazing land and cash. Sorghum, cowpea, sesame and mungbean were included in the game as crops that participants could choose to grow on their plots. The resources of the game's four participants differed according to the four existing farm types (Assogba et al., 2022). The participants with a SOC, SOL, MOD or LCL farm had respectively five, four, six and four cultivated fields of the same area. Some of the resources are expressed in their actual unit (e.g. family size = number of household members); whereas other resources such as fertilizer, manure and money had a generic unit. The resources at the beginning of each game session correspond with the average characteristics of each farm type (Table 2).

2.2.3.3. Dynamics. Crop production in the study area occurs during the rainy season and is a function of rainfall, manure, fertilizer and mulch. Therefore, we assumed that crop production increases with organic

Table 1
Game development steps from the prototype of the game and the final version used in this study. The numbers in parenthesis refer to the number of participants.

Date and activities	Participants	Institutions	Update of the game
12/2020: one test session	Researchers (6)	Wageningen University (WU), Centre de coopération internationale en recherche agronomique pour le développement (CIRAD)	Reduction of the steps in the game from 12 to 6.
01/2021: two tests sessions	Researchers (6)	WU, CIRAD	Simplification of rules on crop and livestock production as well as prices. Reduction of steps in the game from 6 to 3.
02/2021: one test session	Researchers and local agronomists (10)	WU, CIRAD, Université Joseph KI-ZERBO (UJKZ), Associaion Minim Song Panga (AMSP)	Game play adjustment: repartition of roles to facilitators. Adjustment of rules on inputs in fields and livestock selling: maximum 2 organic and mineral inputs per field. Unfed animals cannot be sold.
03/2021: two tests sessions	Farmers (4)		Removal of rules on land cultivation according to labour availability. Adjustment of rules on free grazing on crop residues.
04/2021: two tests sessions	Farmers (4)		Introduction of rules on seasons (rainfall) variability. Each season could be "good" or "bad".
04/2021: debrief meeting	Researchers (5)	WU	Fixed sequence of seasons for easier comparison between game sessions.
06/2021: conceptual update of the game	Researchers (5)	WU	New design of the game to include more than one crop and intercropping. Adjustments on crop production rules in relation to the type of season.
07/2021: four tests sessions	Farmers and local agronomists (9)	CIRAD, AMSP	Adjustment of prices and participant resources at the beginning of the game.
08/2021: two tests sessions	Farmers (4)		Adjustment of rules on manure production by livestock.

Table 2
Resources allocated to each farm type at the beginning of a game session. SOC = subsistence-oriented crop; SOL = subsistence-oriented livestock; MOD = market-oriented and diversified and LCL = land constrained and livestock farms. Fertilizer, grains, crop residues, manure and money have a generic unit whereas the other resources are expressed as the actual number.

	Family size	Cows	Donkeys	Sheep	Poultry	Fertilizer	Grains	Crop residues	Manure	Money
SOC participant	5	0	0	2	0	0	0	0	0	1
SOL participant	4	0	0	4	0	0	0	0	0	1
MOD participant	6	2	1	2	0	0	0	0	0	2
LCL participant	4	5	0	2	0	0	0	0	2	2

(manure and mulch) and mineral inputs, but not to an unlimited extent (Gram et al., 2020). Organic inputs also limit soil fertility decline and support crop yield (Adams et al., 2020; Soma et al., 2018). Information on crop production was included in the mechanics and the rules of the game (2.2.4). In each round (2.2.4), fields in the game produced grain and crop residues as a function of rainfall and inputs (manure, fertilizer and mulch) supplied by the participants. Rainfall was represented by the type of season which could be "good" or "bad". A "good" season referred to adequate and well-distributed rainfall throughout the season whereas a "bad" season referred to the opposite and implied a yield penalty (see Box S1). To simplify the game, labor was not considered for crop production. However, labor was considered to determine the maximum amount of crop residues a participant can harvest (see Box S1). Yield loss due to prolonged monoculture (Bado et al., 2012) was included in the game as grain loss. In addition, benefits from cereal-legume intercropping and rotation (Franke et al., 2018; Ganeme, 2022) were translated in a grain bonus in the game (Box S1). Two types of intercropping were represented in the game, namely the traditional intra-hill intercropping (only for sorghum-cowpea intercropping) and strip intercropping (two lines of sorghum alternated with two lines of another crop).

Livestock production in the study area depends on grazing in rangelands that are common property of the communities (Assogba et al., 2023) and only produce grasses during the rainy season. Similarly, the grazing land in the game produced forage during the sowing period only. It was assumed that livestock could survive on the amount of forage produced during that period (Assouma et al., 2018). To simplify the game, we assumed that the grazing areas within the study area were able to provide enough forage for livestock in the rainy season. In addition, we assumed that only livestock belonging to the study area could graze on grazing areas. The movement of agro-pastoralists' livestock outside the village were not considered and their herd size was adjusted to reflect observed data (Table 2 and Table S1). The contribution of trees and shrubs to livestock feeding in the dry season was not considered for simplicity. According to (Otte and Chilonda, 2002) the calving and lambing of cattle and small ruminants in mixed farm systems of SSA happens on average every two and one year respectively. In the game, livestock reproduction was simplified and only occurred at one moment (Box S1). Manure production from livestock increases with their weight (Assouma et al., 2018). In the game, a larger amount of manure is produced by cattle followed by donkeys and small ruminants. Manure production in the game happened once each round (Box S1) and for each participant manure produced on farm by their livestock was automatically stocked in their farm and could later be applied to fields.

Although prices of resources varied within and across years in reality, they were kept constant in the game to simplify the playing process. Nevertheless, prices in the game reflected general patterns in the study area. For example, cattle were the most expensive livestock and cowpea had a higher price than sorghum.

2.2.3.4. Interactions. Interactions between farm households and resources were implemented in the game (Diarisso et al., 2015). Participants invested labor, fertilizer and manure in their fields and harvested grain and crop residues that were used to feed their family and livestock respectively. Participants could also buy and sell livestock as well as grains and crop residues. In the game, game participants were also free to discuss and exchange biomass (manure, crop residues, grains and livestock), fertilizer and lend money to each other.

2.2.4. Game mechanics

The rules of QUEEN are listed in Box S1.

The game was played for five consecutive rounds in our study but it could be played for more rounds if needed or desired. Each round of the game corresponded to one year. Each round consisted of three steps: sowing, harvesting and trading. Game sessions would start with an explanation round to help participants become familiar with all the rules of the game and ask questions. Four facilitators were involved in the game. Two updated crop and livestock production in the game, one reported statistics, and the last one bought/sold resources from/to participants.

In the first step of the round (*sowing*), each participant chose the crops they wanted to grow. Participants could grow a maximum of two crops (intercropping) in a field. Participants could choose to apply intercropping and crop rotation, as well manure and/or fertilizer on each field. Fertilizer and manure could be bought if participants had enough cash. During the sowing step livestock were grazing on the grazing land, where they always met their feed requirement.

After all participants completed the first step, one facilitator announced the type of season which could be "good" or "bad" (see Dynamics section of 2.2.3). A series of seasons was randomly chosen and used in all sessions. The season was always "good" for the first, fourth and fifth round while it was "bad" for the second and third round. Likewise, the same initial and external conditions were used for all participants in every session.

In the second step (*harvesting*), participants harvested all grains produced according to the inputs applied and the type of season. Each participant harvested the crop residues according to the available labor, determined by the family size, and the number of cows and/or donkeys (Box S1). All animals in the game could then graze freely on crop residues left on the fields, as commonly practiced in the study area.

The third step (*trading*) was dedicated to buying and selling at the market. We assumed unlimited resources availability in the market and constant prices throughout sessions (Box S1). This meant that for every seller there was a buyer and vice versa. Participants could also exchange resources among themselves and they were free to decide on the rules of exchanges. Participants who could not meet their feed/food requirement could purchase additional crop residues/grains to feed their livestock/family. Unfed animals could not be sold.

2.2.5. Game solution space

It is possible to numerically explore serious games through the solution space determination (Speelman et al., 2014). The solution space of a game can be defined as all the possible outcomes of the game. By projecting the outcomes of game participants in the game solution space, one can reflect on the participants' performance vis à vis the potential outcomes of the game.

To explore the solution space of QUEEN, a digital version of the game was created in Netlogo 6.2.1 (Wilensky, 1999). The physical board game was transformed into an agent-based model in Netlogo. The digital version included four "farmer" agents whose decision-making can be automated or controlled by a human. In order to explore the range of

possible outcomes offered by QUEEN, a routine was set up to allow automated agents to play for a given number of rounds. In total 30,000 sessions (one session = five rounds) were ran with four automated farmer agents. We assumed 30,000 automated sessions would be enough to capture a wide range of potential outcomes of the game in terms of food security, cash, and crop and livestock production. That choice was based on five reasons:

1. The relatively high number of choices in resources' management in a farm (e.g. 210 combinations of crops possible with four fields).
2. The relatively high possibilities of exchange/buying/selling among participants and between participants and the market. Each participant being able to exchange/buy/sell amounts ranging from the minimum to the maximum of resources they have.
3. The computation time required for the 30,000 simulated sessions (12–15 h).
4. The computation time required to process and analyze the results of the 30,000 simulated sessions.

Therefore, the choice of the 30,000 simulations was a compromise between the possible outcomes of the game and computation time required for the simulations and the analysis of their results. Given that agents in the solution space exploration were taking random decisions, the resulting solution space was probably an approximation of the real one. We opted for random decision-making because of simplicity of implementation and computer limitations as opposed to search (evolution) algorithms that would require larger numbers of model runs and are often computationally expensive (Chérel et al., 2015; Kumar et al., 2010).

Initial resources (Table 2) allocated to each agent were the same as the ones used during the game sessions with farmers. Random decision-making was made by each automated participant on crop and livestock management, buying and selling of resources as well as exchange of resources.

2.3. Game sessions

Three pilot game sessions were organized, using the final version of the game, on the 17, 18 and 19 of November 2021 in Yilou (one session) and Tansin (two sessions). Each session lasted on average 2.5 h and involved four participants, resulting in a total of 12 unique participants. Each participant took part in only one game session. All participants were previously included in a database containing data on the resources of their farms (Assogba et al., 2022). Each participant played a farmer whose resources corresponded to the farm type the participant belonged to. Because the LCL farm type was only present in Yilou, in the two game sessions that took place in Tansin, LCL farms were replaced by SOC farms, which is the dominant farm type in Tansin. At the start of each session, participants were reminded of their objective in the game which was to increase as much as possible, crop and livestock production as well as cash.

2.3.1. Data collection

2.3.1.1. During the game. Throughout the game sessions, we collected data on participants' cropping system and livestock management decisions as well as exchange of resources among the participants and with the market in every round (monitoring sheets, supplementary materials). Participants decided which crops to grow, how much fertilizer, manure and/or mulch to apply, and if they intercropped or rotated their crops. Grains and crop residues harvested, purchased, sold and/or exchanged were also recorded. Livestock (cattle, sheep and donkey) sold, purchased and dead from starvation were also recorded. We also kept track of the number of household members that could not be fed and the cash each participant had at the end of each round.

2.3.1.2. After the game. After each game session, a post-game questionnaire (supplementary materials) was used to collect information on participants' experience during the session. The post-game questionnaire was used right after each game session and aimed at collecting data on: (1) participants' objectives during the game and if and how they achieved them or not and why, (2) actions implemented during the game and how these differed from reality, (3) how participants influenced each other during the session, and (4) what participants learnt and their recommendations to improve the game.

2.3.2. Data analysis

The data from each game session was analyzed per session, per round, per participant and per farm type. Average values were computed for resources per farm type, except for the single LCL participant who was involved in the sessions.

Based on the data collected with the questionnaire after each game session, we qualitatively described the diversity of participants' objectives and computed the proportion of participants using a given strategy. A strategy was an ensemble of implemented actions by participants to reach their objectives. To analyze the degree of realism of participants' decision-making during the game sessions, we calculated the proportion of participants that reported they would have used a different strategy in real-life and we analyzed what they would have done differently. In addition, we qualitatively assessed the realism of the game itself based on participants' feedback on what was missing in the game in comparison to real-life. Moreover, we summarized what participants reported they learnt from the game sessions.

From the solution space exploration, Pearson correlation tests were computed to assess possible trade-offs or synergies between simulated outputs of the game, such as the amount of cash and the number of food insecure household members. Correlation tests were computed for the last round of the game per farm type and for all participants. In addition, probabilities of occurrence of the simulated game output values indicated the likelihood of achieving a specific amount of each resource (cash, cattle, donkey, small ruminants, grains and crop residues).

Moreover, in order to categorize the crop and livestock production levels of participants, the solution space was divided into nine zones (Fig. 3). The zones were based on cash equivalent of crop and livestock production in the fifth round of the game. Three levels of production were defined for crop and livestock each, based on the first and the third quartile of values from the solution space:

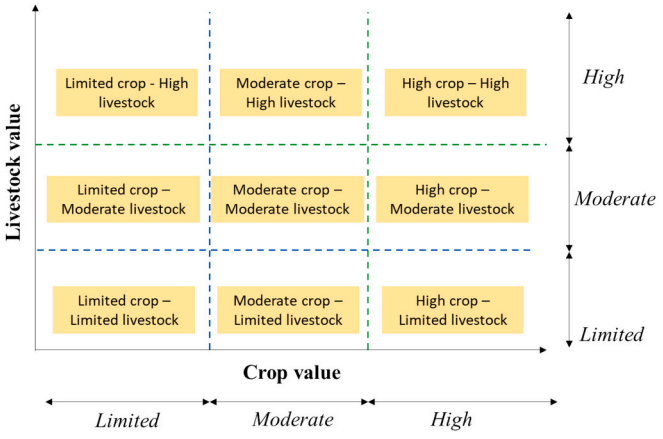


Fig. 3. Stylized representation of the nine crop-livestock zones of the solution space based on crop and livestock value (i.e. cash equivalent) in the fifth round of the game. The blue and the green lines are respectively the first and the third quartile. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

1. Crop/livestock value (i.e. cash equivalent) lower or equal to the first quartile was considered "limited"
2. Crop/livestock value (i.e. cash equivalent) greater than the first quartile and lower or equal to the third quartile was considered "moderate".
3. Crop/livestock value (i.e. cash equivalent) greater than the third quartile was considered "high".

The combination of the crop and livestock levels of production led to nine zones reflecting different performances of mixed crop-livestock systems (Fig. 3).

The outcomes obtained by the game session participants were projected in the simulated solution space. We then checked whether and how participants' outcomes fitted in the solution space. Moreover, based on the game objectives of increasing crop and livestock production and cash, we defined the optimal simulations as those with cash, crop and livestock related output values comprised between 0.75 and 1.25 times the 95 % percentile value from the solution space. An additional requirement for the optimal simulations was that all households' members received the required amount of food during each of the five rounds. The strategies and outcomes of the real-life game sessions and the optimal simulations outputs were compared.

3. Results

3.1. Game session results

3.1.1. Farm management during game sessions

The cropping system during game sessions was dominated by sorghum, occupying on average 45 % of plots (Fig. 4). The second most cultivated crop was cowpea (34 % of plots) followed by sesame (17 % of plots). Cowpea and mungbean were often (in 63 % of all plots) intercropped with sorghum using the strip intercropping practice. Crop allocation to plots remained constant for most participants throughout rounds but varied among participants without a clear relation to the farm type. The traditional sorghum-cowpea intercropping was applied only in 5 % of plots on average. Market-oriented (MOD) participants had the highest number of plots and also had the most diversified cropping system. The agro-pastoralist (LCL) participant, cultivating the smallest number of plots (together with SOL), had the least diversified cropping system which consisted almost only of sorghum and cowpea. The newly introduced mungbean crop was rarely included by participants in their cropping system.

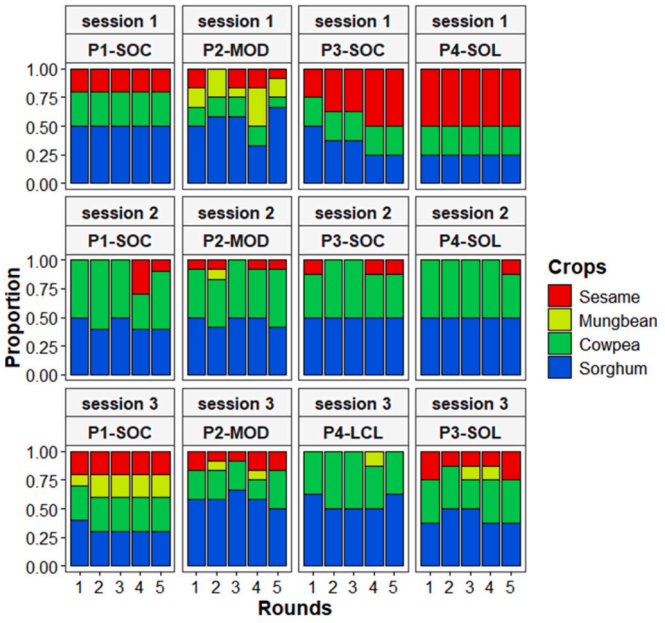


Fig. 4. Plots allocation to crops. SOC = subsistence-oriented crop; SOL = subsistence-oriented livestock; MOD = market-oriented and diversified and LCL = land constrained and livestock (LCL). P1, P2, P3 and P4 refer to the four participants of a game session. Year 2 and 3 are "bad" years.

Manure and fertilizer were frequently applied by participants in each round, respectively 59 and 45 times over 60 possibilities across all three sessions (Fig. S2, supplementary material). Mulch application occurred only 29 times. Fertilizer and organic (manure and mulch) inputs were mostly applied to sorghum (Fig. S2, supplementary material). On average, 56 units (over 190 units possible) of organic (manure and mulch) inputs and 23 units (over 190 units possible) of mineral fertilizer were applied per game session (Table S2, supplementary material).

Participants mainly purchased grains in bad years when the harvests were low, i.e. in rounds two and three (Fig. 5a). Only sorghum grains, having the lowest price, were purchased as additional food to feed the family. Subsistence farmers (SOC and SOL participants) in the first session had a cropping system dominated by sesame and purchased grains more frequently. However, they also sold most grains (especially sesame) per capita. For all participants, grain selling occurred mainly in the good years, especially in rounds four and five, and for the most part concerned cowpea and sesame.

Livestock purchase and sale were rare compared to grains (Fig. 5b). Contrary to grains, livestock was largely sold in bad years. Small ruminants were the most frequently sold animal type. Similarly, livestock purchase mostly concerned small ruminants and occurred in the last (fourth and fifth) rounds of the sessions.

The purchase of crop residues was frequent in the first game session, especially by SOC and SOL participants who had a crop system dominated by sesame which crop residues could not be grazed by livestock (Fig. S3, supplementary material). Additionally, MOD (in sessions 1 and 3) and LCL (in session 3) purchased crop residues mostly in bad years. Crop residue sales were rare and occurred the most in the fourth and fifth rounds.

3.1.2. Participant strategies during game sessions and their perceptions of the game

When asked at the end of the game session, participants' objectives differed (Table S3, supplementary material). Actually, while some were trying to increase their cash and/or crop/livestock production, others were trying to simply maintain their resources (cash, grain, manure, crop residues, and livestock) or simply feed their family. Nine participants out of 12 involved in all sessions said they reached their objective in the game. The strategies participants used to reach their objectives also varied. The use of strip sorghum-legume intercropping was the main strategy reported with eight participants reporting they used strip sorghum-cowpea/mungbean intercropping to reach their objectives. In addition, five participants reported grain selling as their strategy to get cash.

Participants reported differences between what they did in the game sessions and what they actually do in real life. Three participants reported not selling grains, especially sorghum grains in real life. In addition, in the game, some participants increased the share of land dedicated to sesame beyond what is really done as four of them reported focusing much more on sorghum and cowpea in reality. Two participants reported they would have applied the traditional intra-hill sorghum-cowpea intercropping in their fields rather than the strip intercropping they used in the game. With respect to crop residues management, most participants agreed on harvesting a higher share of crop residues in reality than what happened in the game, as the rules of the game did not always allow to harvest the total amount of crop residues produced. While 11 out of 12 participants applied mulch at least once in their fields in the game sessions (Fig. S2, supplementary material), only three participants (2 SOC and 1 SOL types) reported using crop residues as mulch in real life. Free grazing on crop residues left on fields after harvest in the game appeared normal for most (9 out of 12)

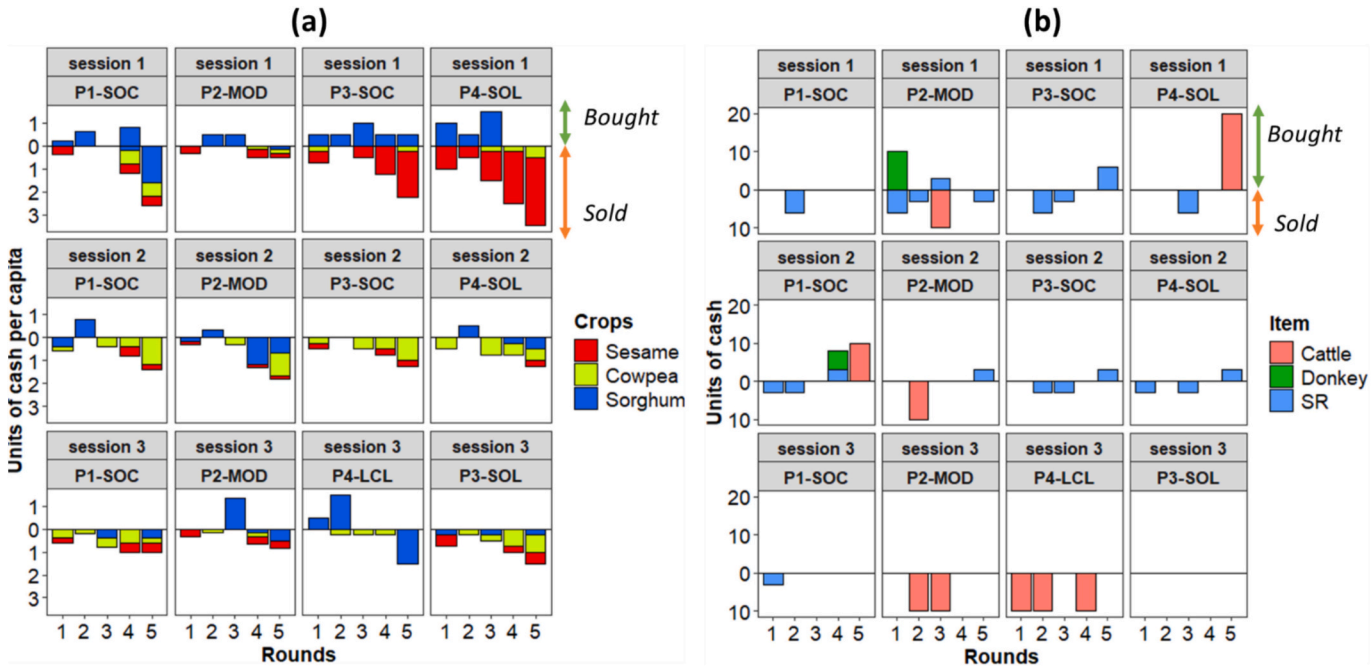


Fig. 5. Values of grains per capita (a) and livestock (b) purchased and sold during game sessions. SR = small ruminants. SOC = subsistence-oriented crop; SOL = subsistence-oriented livestock; MOD = market-oriented and diversified and LCL = land constrained and livestock (LCL). Year 2 and 3 are "bad" years.

participants. Participants reported that the real-life manure benefit of livestock grazing their fields was not included in the game. Interestingly none of the participants reported selling livestock as part of their real-life strategies. Even if most participants' reported after the game that they exchange seeds, grains and crop residues in real life, only three cases of biomass exchange were noted between participants in the sessions. Indeed, during the game sessions, exchange of items between participants were rare and only concerned crop residues given away (two occurrences) and exchanged for money (one occurrence).

Concerning "lessons" learnt from the game session, participants mainly reported they learnt about the benefits of intercropping especially in bad seasons as continuous intercropping allowed extra yields in good and bad seasons in the game (Box S1). Some participants also reported they learnt about farm management in bad seasons without being specific about it. Only one participant reported learning about crop rotation despite crop rotation being applied by all participants during the sessions.

3.2. Game solution space

The exploration of the solution space showed that the game allowed a wide range of outcomes (Fig. 6). Most simulations (30 %) were in the "moderate crop – moderate livestock" zone. Few simulations (5 %) resulted in high livestock production combined with limited crop production. Likewise, the game offered a relatively small probability (7 %) of high crop production with high livestock production. There was no clear relation between the possible crop-livestock systems and the total number of food insecure household members in a game session (Fig. S4, supplementary material), because the latter strongly depends on the decision of a participant concerning grain selling (which can be unrealistic). Similar to the number of food insecure household members, participants' cash in the fifth round of the game was independent from the performance of crop and livestock production (Fig. 6).

The analysis of the solution space revealed a weak and negative correlation (correlation = − 0.1, *p*-value <0.001) between the number of food insecure household members and the amount of grains harvested (Fig. S5, supplementary material). Indeed, feeding household members with purchased grains, and/or remaining food insecure irrespective of

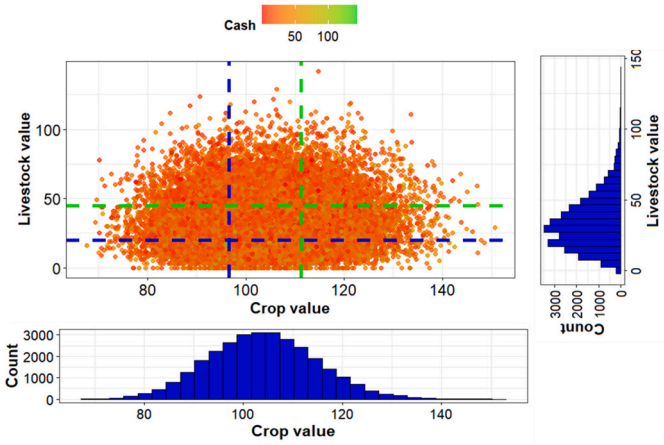


Fig. 6. Crop and livestock values expressed in cash unit achievable with QUEEN. The blue and the green lines represent respectively the first and the third quartile of crop/livestock values from the partial solution space. Blue bars are histogram from crop and livestock values in the solution space. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the amount grains available was possible in the game, even though the latter was not a realistic option. In addition, the amount of cash owned at the end of the game was weakly associated to grains and livestock selling/buying (correlation = 0.05–0.1, p-value <0.001). This was explained by the possibilities of exchange between participants (Fig. S6a, supplementary material) and the use of their cash to buy/sell resources (e.g. fertilizer) during the game. Furthermore, the solution space exploration showed that a limited number (maximum four to six) of cattle could be maintained by each participant at the end of the game. However, the maximum number of cattle for each participant (four to six) was only observed 17 times during the exploration corresponding to only 1/10000 chance of occurrence. In contrast, it was much more likely (99.9 %) for each participant to end up with between zero and two cattle at the end of the game. The solution space exploration also showed that

it was not possible to have both a relatively high number (from four to six) of cattle and a lot of cash, crop residues or other livestock (small ruminants and donkey). The same incompatibility with cattle applied for donkeys and small ruminants. For example, the average cash, small ruminants and donkeys owned by each game participant with a number of cattle greater than four was respectively: 5.2 cash units (min = 0 and max = 80), 1 small ruminant (min = 0 and max = 20) and 1.6 donkey (min = 0 and max = 15).

3.2.1. *Participants' outcomes in the game solution space*

The crop values of simulated agent farmers were always higher than those of real-life participants' (122 and 94 units of cash on average for optimal simulations and game sessions resp.) (Fig. 7). On the opposite, the average livestock value of the game sessions was superior to that of the optimal simulations (77 and 65 units respectively). Moreover, the game sessions yielded similar cash with the optimal simulations (on average 58 and 54 units of cash resp.). Furthermore, the total value by combining crop, livestock and cash was lower in game sessions compared to optimal simulations (229 and 241 on average for game sessions and optimal simulations resp.).

Interestingly, across game sessions and throughout the rounds of each game session, real-life participants remained mainly within the "Limited crop – High livestock" production zone (Fig. 8a). This suggests that the maintenance of high livestock production in the game was key for real-life participants. Likewise, crop production remained limited/moderated but varied more strongly from one round to another compared to livestock production. Contrary to real-life participants, agent farmers in optimal simulations progressively moved from limited crop production and high livestock production to high crop production combined with high livestock production (Fig. 8b).

3.2.2. *Farmer strategies versus optimal simulations*

The cropping system in the optimal simulations was different from the game sessions (dominance of sorghum and cowpea) as plot allocation was similar for each crop (Fig. 9a). The crop allocation to plots remained mostly the same throughout the rounds even if the share of intercropped sorghum-legume plots increased. This contrasted with real-life participants whose cropping system was dominated by sorghum and cowpea (Fig. 4).

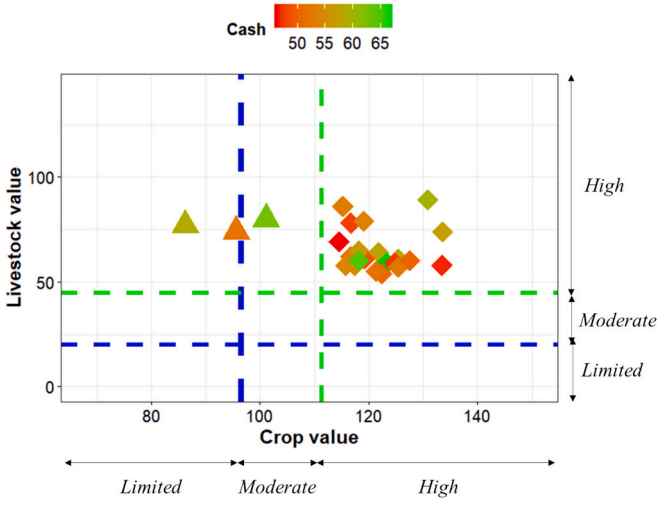


Fig. 7. Projection of each game session's outcome in the solution space (triangles). Optimal simulations of the solution space regarding crop, livestock production and cash (diamonds). The blue and the green lines represent respectively the first and the third quartile of crop/livestock values from the solution space. Crop and livestock values are expressed in unit of cash. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Mineral fertilizer was mainly applied to sorghum whereas organic fertilizer (manure and mulch) was mainly applied to legumes in optimal simulations. The total amount of fertilizer applied in the optimal simulations was always higher than during the game sessions (Fig. 9b). Whereas the total amount of manure applied in the optimal simulations was similar to the one observed in game sessions, the amount of mulch applied in optimal simulations was much higher than the game sessions. In the game, applying mulch resulted in additional grain and the price of crop residues for mulch was only half of that of fertilizer and manure. Therefore, by applying one unit of residue (0.5 unit of cash) as mulch it was possible to get one to two units of cash depending on the cultivated crop. In comparison, applying one unit of manure or fertilizer (1 unit of cash), resulted in a crop value of 1.5 to 2.5 units of cash. Hence, the cost-benefit ratio of mulch (0.25–0.5) was much more beneficial than that of manure/fertilizer (0.4–0.67).

In optimal simulations, manure, fertilizer and crop residues were frequently purchased (Fig. 10a). Additionally, agents sold cattle to purchase donkeys in the first rounds. Donkeys had lower feed requirement but provided similar labor to cattle in the game. This allowed agents to decrease pressure on the cropping system while maintaining the same labor availability in the game. However, simulated selling was in general less frequent than in the game sessions (Fig. 5) and mainly concerned grains harvested. In the optimal simulations, many exchanges of resources occurred between agent farmers in each round (Fig. S6a, supplementary material) whereas in the game sessions these were rare. The number of items exchanged increased with the number of rounds. On average, manure, fertilizer, grains and crop residues were given and received by SOC and SOL agent farmers respectively. (Fig. S6b, supplementary material). In the last round of the optimal simulations, almost no cattle remained (Fig. 10b). Cattle were mostly replaced by donkeys, which was not the case in the game sessions in which participants maintained cattle as much as they could (data not shown).

4. Discussion

We developed the first serious game which considers all components of mixed crop-livestock systems in semi-arid West-Africa. The development of the game was an iterative process involving farmers and researchers with the ambition to find a good balance between easiness to play and capturing the key elements driving system performance (Speelman et al., 2017). The game was conceived to serve as a virtual laboratory and allow discussion around individual farm management and communal management of resources (e.g. crop residues after harvest). Furthermore, we explored the solution space of the game and analyzed the game's outcomes and trade-off between components of the farming system (e.g. crop, livestock) of as represented by the game. In what follows we first discuss the degree of realism, participants' learning, the outcomes of the game as influenced by its mechanics and what this means for messages conveyed through the game. We end with a discussion on further use and improvement of the game.

4.1. *Assessment of participants' behavior and game realism*

Based on debriefing sessions organized with participants just after each session, we tried to better understand their choices in the game and what they learnt from it (Andreotti et al., 2020). Additionally, data collected during the debriefing sessions made it possible to put the participants' choices during the game sessions into the perspective of real-life management of a farm. This approach allowed to analyze how participants' choices were related to the simplification of socio-ecological processes in the game.

Clarifying participants' objectives after each game session, even if the generic objective of the game was announced to all participants before each session, proved to be key in understanding participant choices in the game. Most participants involved in game sessions did not try to simultaneously increase cash, crop and livestock production,

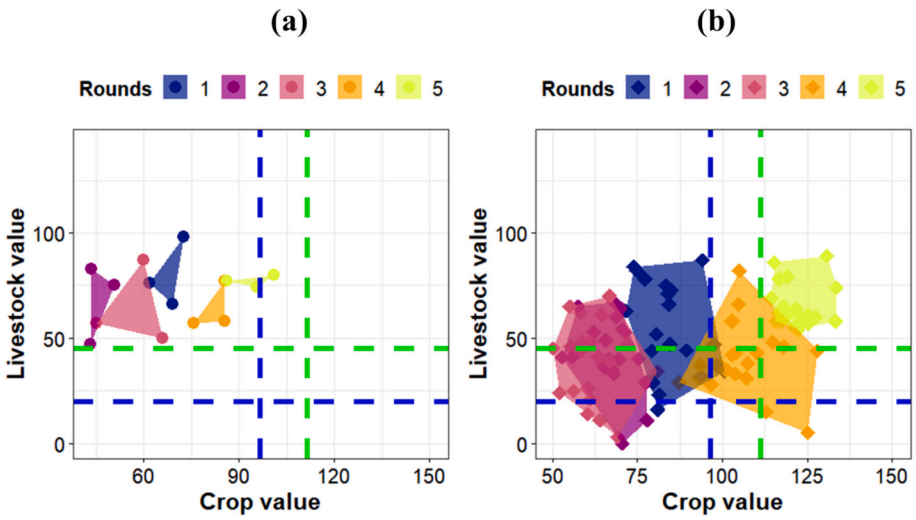


Fig. 8. Crop and livestock production value (unit of cash) throughout rounds in game sessions (a) and optimal simulations (b). The dots are game sessions or optimal simulations outcomes. Each polygon represents an ensemble of optimal simulation or a game session outcome for a given round. The blue and the green lines represent respectively the first and the third quartile of crop/livestock values from the solution space. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

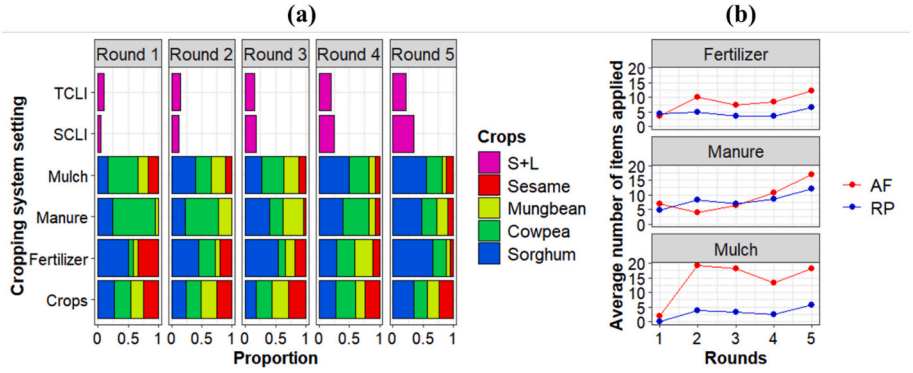


Fig. 9. Plot, organic and mineral fertilizer allocation to crops, from optimal simulations in game solution space (a). Comparison between average inputs (fertilizer, manure and mulch) applied by all participants in game sessions by real-life participants (RP) and agent farmers (AF) (b). S + L = Sorghum and Legumes (cowpea/mungbean) intercropped. SCLI = Strip Cereal-Legumes Intercropping. TCLI = Traditional Cereal-Legume Intercropping.

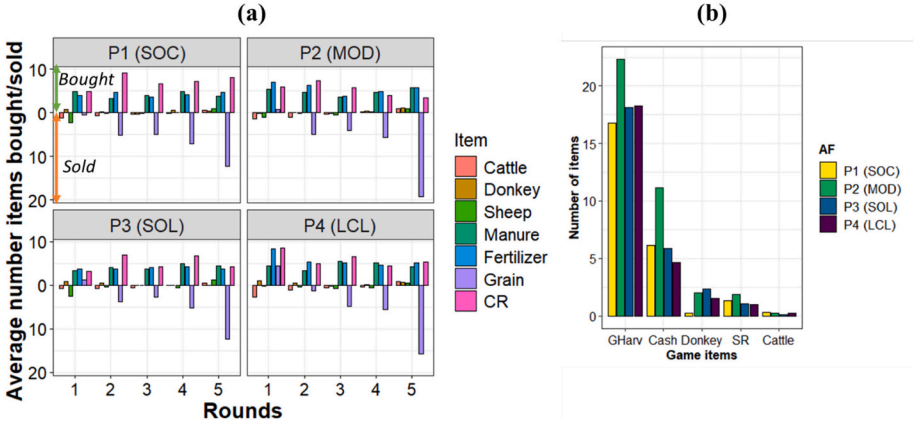


Fig. 10. Patterns of optimal game simulations. Average number of items bought and sold (a). Average number of items owned at the end of optimal simulations (b). P1, P2, P3 and P4 are agent farmers (AF). AF farm type: SOC = subsistence-oriented crop; SOL = subsistence-oriented livestock; MOD = market-oriented and diversified and LCL = land constrained and livestock (LCL).

which was the generic objective of the game. They often tried to increase one or two elements (e.g. crop production and cash) without necessarily trying to maximize them. For example, it was possible for real-life participants to apply more crop inputs (e.g. fertilizer) than they did, and thereby increase crop production. However, the participants did not systematically use their cash to purchase crop inputs or resources but rather spent as little as possible and saved cash. This may be linked to participants' individual objectives which were in general to increase their resources, and especially cash. But it could also be explained by real-life risk-aversion of participants who live in an uncertain environment in terms of climate and security (Huet et al., 2020).

The questionnaire applied at the end of the game sessions revealed the level of realism of participants' choices during the sessions (Sari et al., 2024). Whereas participants' farm management in the game was close to reality (Assogba et al., 2023; Ganeme et al., 2021) their decision-making was also influenced by rules embedded in the game's mechanic as they tried alternative management such as strip intercropping. This suggests that the game encouraged participants to try new management options they would otherwise not apply in real-life. Because serious games are not only simplification of reality but can also be used for exploration of socio-ecological systems, the ability of a game to make participants think beyond the usual management and try new options is essential (Speelman et al., 2017). Furthermore, participants' feedback suggested a satisfactory level of realism of the game. This does not mean that the game precisely described outputs of the systems (e.g. yields) but rather that the key socio-ecological mechanisms of the farming system were represented (Garcia and Speelman, 2017). The satisfactory level of realism reported is most likely the fruit of the development process of the game which allowed to capture researchers' and farmers' perceptions of the farming system functioning. A satisfactory level of realism offers the opportunity to build trust in the outcomes of the game and move beyond current management to explore alternative ones (Moojen et al., 2022).

In "open" serious games such as QUEEN, where interactions among participants are not strictly predefined, the possible emergence of collective "solutions" for the management of socio-ecological systems is an advantage (Andreotti et al., 2020). Open serious games do not have a pre-defined set of solutions and have the advantage of letting participants express their creativity within the boundaries of the game. However, no collective management of resources was observed as participants were mostly focused on achieving their objectives alone. Even though there were possibilities of exchanging items, participants' interactions were more verbal and direct exchange of items remained limited. The lack of direct biomass (e.g. crop residues and manure) exchange between farms, especially livestock and crop-oriented farms, has already been reported in the study area (Assogba et al., 2022). As farms do not produce enough feed for their livestock and grain yields are modest with small organic and mineral inputs, the possibilities of exchange between farms are limited. Nevertheless, the solution space exploration showed that a considerable exchange between farms was possible even if these exchanges would result in significant changes in the structure (e.g. herd size) of participants' farms. Hence, although in the game it was possible to try new farm systems settings (e.g. moving towards more specialized farms), participants remained close to their real-life farm management and farm types.

Participants' learning from serious game sessions is often assessed through pre-game and post-game surveys to gain insights in the key messages they obtained from the game (Bosma et al., 2020; Salvini et al., 2016). In this study, information on the farming system characterization was already collected through a previous household survey and farm monitoring (Assogba et al., 2023; Assogba et al., 2022). Hence, only the post-game survey was applied to determine participants' learning. Participants learnings from our game were mainly related to: (1) the functioning of the farming system, and (2) the alternative farming options included in the game. However, what participants reported they learnt from the game might not necessarily be applied in real life but could only be applicable in the game as a strategy to reach their objectives in game sessions, especially if they mentioned the second learning outcome. As an example, many participants reported they learnt about the advantages of (strip) intercropping. But they also reported that they would have stuck to the traditional intercropping in real-life. This is most likely due to additional labor and financial investment related to strip intercropping (Kermah et al., 2017), making this practice less feasible and risky. Therefore, as we did not have prior data on participants' perspectives on the farming system functioning, learning outcomes from the game may be biased. Moreover, similar to this study, many studies also reported learning about agricultural practices achieved through serious gaming in various socio-ecological contexts (Bosma et al., 2020; Den Haan and Van der Voort, 2018; Sari et al., 2024). However, whether such learning outcomes will translate into real-life changes often remains unknown because of the influence of other system drivers such as policy and market dynamics. Ideally, learning from serious games should be followed by real-life experiments (Lairez et al., 2020) which in return would generate information to adjust the game to continuously reshape and create knowledge.

4.2. Game solution space exploration

Serious games are models (Villamor et al., 2023), and models are based on assumptions and simplifications. Therefore, analyzing the extent to which assumptions and mechanisms influenced game outcomes is relevant for a better understanding of the latter. We ran a solution space exploration and analyzed the range of outcomes of the game to better understand possible outcomes of the game in relation to its mechanisms and assumptions.

The solution space exploration showed that the game incorporated the complexity of the farming systems allowing a vast diversity of farm management options and interactions among participants. The diversity of outcomes and associated probability of occurrence could be analyzed with the solution space. In addition, the solution space analysis helped understanding key game mechanisms (e.g. buying, selling and exchanging) influencing outcomes of the game. Moreover, we were able to identify trade-off between the main outcomes (crop, livestock and cash) of the generic objective of the game. Therefore, the solution space analysis can significantly contribute to the process of game development. In fact, the process of building game is often an iterative one with multiple tests with researchers and stakeholders to guarantee the playability and the credibility of the game. However, even with a high number of tests there may still be management options in the game leading to undesirable outcomes. For now, the solution space analysis is what is closest to a sensitivity analysis (Borgonovo et al., 2022) of serious games which can help build more credibility in serious games.

The solution space exploration helped in identifying theoretical "strategies" leading to outcomes that satisfied multiple objectives. These "strategies" were similar to participants' management regarding manure and fertilization application as well as grains selling as the main way to get cash. However, the theoretical farm management "strategies" suggested by the optimal simulations of the solution space implied a redesign of the current farm system that comprised (1) increasing mulch and the share of land allocated to legumes; (2) decreasing the number of small ruminants and cattle; (3) replacing cattle by donkeys to lower feed requirements of the herd while maintaining labor availability and (4) buying crop residues for feed and mulch. The results of the optimal simulations depend on the simplifications made in the game to facilitate its playability. In real-life a significant decrease in the area allocated to sorghum would decrease the amount of grains and crop residues produced to feed the household and the herd, whereas this is not the case in the game where all crops produce the same amount of grain and crop residues. In fact, the important role of cereals supporting food self-sufficiency and livestock production has been demonstrated in semi-arid West-Africa (Falconnier et al., 2017; Huet, 2023). Another assumption that explains the "strategies" of optimal simulations of the

solution space is the unlimited availability of resources, and particularly crop residues, in the market. In reality, crop residue production is limited in semi-arid Burkina Faso and only a small fraction of crop residue produced is sold by farmers (Assogba et al., 2022). Therefore, the massive purchase and application of crop residues as mulch is not a realistic scenario. Additionally, mulching would only be effective if farmers could fence their fields (which cost is not taken into account in the current version of the game) which is unlikely with their limited livelihoods and would also increase feed scarcity for large livestock owners (Andrieu et al., 2015; Berre et al., 2021). In the livestock subsystem, decreasing herd size, especially in crop oriented farms, is theoretically possible but due to multiple services and the social value attached to livestock keeping (Descheemaeker et al., 2016; Molina-Flores et al., 2020), especially cattle, the feasibility remains limited. Therefore, even if the optimal simulations were theoretically possible and were on many points similar to participants' management, their implementation in reality is difficult.

4.3. Further use and improvements of QUEEN

The present study focused on the design of the game, the results of a few (pilot) sessions with farmers, and the solution space exploration. Beyond these steps, a serious game can be used for co-design of farming systems leading to practical management plans through scenario definition and analysis. One example is the Dynamix serious game developed by (Ryschawy et al., 2022) for crop-livestock integration in French farms. The authors combined their serious game and a farm model in participatory scenario building sessions, leading to concrete scenarios supported by research.

Further use of QUEEN could explore the effects of intensified collaboration between participants on the farm resources. For example, game sessions can be oriented on knowledge sharing between participants based on their experience. Based on shared knowledge, participants could use the game as a virtual laboratory to evaluate alternative on-farm and off-farm management options. In addition, the game could be explicitly directed towards the exploration of more diverse baskets of options, i.e. an ensemble of options defined by farmers and researchers, from which farmers can choose the most relevant (Ronner et al., 2021). The game can be modified according to specific research questions, for example considering hybrid and/or dual-purpose varieties and related seed prices and crop nutrient requirements. Indeed, participatory breeding is important in the adoption of improved varieties and capacity building of farmers on certified seeds production (Vom Brocke et al., 2020). Therefore, QUEEN could help thinking beyond plant and field level in participatory breeding by stimulating reflections on impacts of improved varieties on the farming system. Concerning nutrient management and livestock movement within and outside villages, QUEEN could be used in association with more quantitative models such as the Agent-based Model of Biomass flows in Agro-pastoral regions of West Africa (AMBAWA) (Berre et al., 2021). The combination of serious games and quantitative agent-based models (ABMs) can potentially address the lack of complex biophysical processes in serious games and the oversimplification of agents' behaviors in ABMs (Villamor et al., 2023). Hence, the game could benefit from biophysical outputs from the ABM while agents' behavior in the ABM could be refined based on participants behavior in the game and their feedback after game sessions.

In the current version of the game, labor (apart from crop residues harvesting) and off-farm income are not included. However, labor is a key factor influencing technology adoption by farmers in West-Africa (Agyekum et al., 2024; Sanogo et al., 2023) and could be included in a future version of QUEEN. Additionally, possibilities of income diversification through off-farm activities could be added to analyze their impact on the farming system. Nevertheless, adding new elements and rules to the game may require simplifications in the representation of the farming system depending on the research question. Indeed, maintaining the playability of the game is key to facilitate communication with farmers.

5. Conclusion

QUEEN is a newly developed serious game for exploring mixed crop-livestock farms of semi-arid West-Africa. QUEEN is based on the local reality of farmers regarding their crops, livestock, and financial status as well as climate. The game design included scientists' and farmers' knowledge on soil fertility management, crop management, livestock mobility and feeding and market prices. QUEEN does not focus on a particular component of the farming system but aims at providing a holistic systems view and analysis. It is a useful tool to foster communication among farmers and between farmers and researchers, because it functions as a virtual laboratory in which farmers are actors rather than spectators and actively share their perceptions and understanding of their farming system. The game triggers participants to consider different components and levels of the farming systems.

The participatory and iterative approach in the design of the game helped to assure a satisfactory level of realism of the game without compromising its easiness to play. However, it was important to clarify participants' objectives and decision-making during game sessions in order to have an objective analysis of the game results. Beyond participatory game development and testing, the exploration of the solution space of the game offered insights into strategies and outcomes achievable in the game. We showed that the exploration of the solution space was relevant to assess the impacts of assumptions and game's mechanics on game's outcomes. The solution space exploration was similar to a sensitivity analysis and therefore can help build trust in serious games as models. Moreover, the realism of strategies to achieve outcomes should be known to researchers. Indeed, our case study showed that the narrative derived from game outcomes needs to be contextualized because the assumptions and simplifications in the game may not always be realistic. Otherwise, messages conveyed by game sessions could be misleading participants. Furthermore, we showed that similar to model exploration, game exploration can be a useful step in the participatory re-design of farming systems. Nevertheless, we recommend that the exploration results need to be discussed with farmers to collectively analyze their relevance.

This paper presents the first version of QUEEN. The game can be improved and adjusted to answer various research questions in mixed crop-livestock systems of semi-arid West-Africa and beyond.

CRediT authorship contribution statement

Gildas G.C. Assogba: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Erika N. Speelman:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **David Berre:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Myriam Adam:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Katrien Descheemaeker:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Funding source

This research was supported by the McKnight Foundation under grant number 17-371.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank farmers, agronomists and researchers from Burkina Faso, France and Netherlands who helped in the design and test phases of QUEEN. Special thanks to Irene Obossou, Nock Ouedraogo, Touwende Sawadogo and Souleymane Sawadogo, Abdoul Aziz Guira, Saiouba Sawadogo and Boureima Sawadogo who facilitated the game sessions. We also think Etienne Delay for providing useful feedback during the development of the game. Thanks to Aminata Ganeme, Abdoul Rasmane Bagagnan, William Lankouande, Louis-Marie Raboin and Jean-Marie Douzet for their inputs during the development of the game. We thank the McKnight foundation which funded this research through the 3F (Feed the soil and Feed the livestock to Feed the people) project and the SESAM (Scenario Evaluation for Sustainable Agro-forestry Management) program. We also thank the CUOMO foundation for providing additional scholarship.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.agsy.2025.104364.

Data availability

Data will be made available on request.

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